

# Quantum Mechanics Cheat Sheet

by Tej

Derived from Leonard Susskind's The Theoretical Minimum (Ch 1..7)  
(Finite-Dimensional Hilbert Space)

## HILBERT SPACE

**Complex Vectors:**  $|\psi\rangle \in \mathbb{C}^n$

**Norm:**  $\|\psi\|^2 = \langle\psi|\psi\rangle$

**Hermitian:**  $A^\dagger = A$ ; eigvals  $\in \mathbb{R}$ ;

◦ eigvecs orthogonal (distinct eigvals);

◦ degeneracy  $\Rightarrow$  simultaneous eigvecs;

◦ Hermitian  $L, M \Rightarrow i[L, M]$

**Unitary:**  $U^\dagger U = I$ , norm preserved

## OPERATORS

**Observable:** Hermitian operator

**Eigen eq:**  $A|\psi\rangle = a|\psi\rangle$

**Expectation:**  $\langle A \rangle = \langle\psi|A|\psi\rangle$

**Commutator:**  $[A, B] = AB - BA$

**SPIN STATES:**  $\{|u\rangle, |d\rangle\}_{\text{basis}}$

$|u\rangle = 1|u\rangle + 0|d\rangle$

$|d\rangle = 0|u\rangle + 1|d\rangle$

$|r\rangle = 1/\sqrt{2}|u\rangle + 1/\sqrt{2}|d\rangle$

$|l\rangle = 1/\sqrt{2}|u\rangle - 1/\sqrt{2}|d\rangle$

$|i\rangle = 1/\sqrt{2}|u\rangle + i/\sqrt{2}|d\rangle$

$|o\rangle = 1/\sqrt{2}|u\rangle - i/\sqrt{2}|d\rangle$

**SPIN OPERATORS:**  $\{|u\rangle, |d\rangle\}_{\text{basis}}$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

**Spin Polarization:**  $\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1$

**Spin Component form :**  $\sigma_n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z$

$\langle\sigma_n \cdot n\rangle = 1$

## STATE - TIME EVOLUTION

$|\psi(t)\rangle = U(t)|\psi(0)\rangle$

**Schrödinger:**  $\hbar \frac{\partial}{\partial t} |\psi\rangle = -iH(t)|\psi\rangle$

**Evolution:**

$\alpha_j(t) = \alpha_j(0)e^{-(i/\hbar)E_j t}$

$|\psi(t)\rangle = e^{-iE_j t/\hbar} |\psi(0)\rangle$  ( $H$  const.)

## EXPECTATION

$\langle L \rangle = \langle\psi|L|\psi\rangle = \sum_\lambda \lambda P(\lambda)$

**Commutator:**  $[L, M] = LM - ML$

**Time evol:**  $\frac{d}{dt}\langle L \rangle = \frac{i}{\hbar} \langle [H, L] \rangle = -\frac{i}{\hbar} \langle [L, H] \rangle$

**Conserved:**  $[Q, H] = 0 \Rightarrow [Q^n, H] = 0$

## SOLVING STATE EVOLUTION

1. Define Hamiltonian  $H$

2. Prepare initial state  $|\psi(0)\rangle$

3. Find eigenbasis of  $H$

4. Express state in eigenbasis:

$|\psi_E(0)\rangle = [E_1, E_2, \dots]^{-1} |\psi(0)\rangle$

5. Expand state:

$|\psi_E(t)\rangle = \sum_j \alpha_j(t) |E_j\rangle$

6. Time evolution:

$|\psi_E(t)\rangle = \sum_j \alpha_j(0) e^{-iE_j t/\hbar} |E_j\rangle$

## SIMULTANEOUS EIG. VECTORS

**Commuting:**  $[L, M] = 0 \Rightarrow$  common eigbasis; simult. diag

◦ Commutation allows simultaneous measurement

## WAVEFUNCTION & BASIS

**Basis:** complete orthonormal  $\{|i\rangle\}$  spans space

**State:**  $|\psi\rangle = \sum_i c_i |i\rangle$ ,  $\sum_i |c_i|^2 = 1$

**Wavefn:**  $\psi(i) = \langle i|\psi\rangle$ ;  $|\psi(i)|^2 = \text{prob.}$

**Basis dependence:**  $|\psi\rangle$  invariant;  $\psi(i)$  depends on chosen eigbasis

**Change basis:**  $c_i = \langle i|\psi\rangle$ ; coeffs change, state unchanged

**Examples:** canonical basis  $\{|0\rangle, |1\rangle\}$ ;

Bell basis  $\frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ ,  $\frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$

## 2D HERMITIAN

$H = a\sigma_x + b\sigma_y + c\sigma_z + dI$

## UNCERTAINTY PRINCIPLE

**Expectation:**  $\langle A \rangle = \sum_a a P(a)$

**Std. dev:**  $\bar{A} = A - \langle A \rangle I$ ; eigenvalues  $a - \langle A \rangle$

**Variance:**  $(\Delta A)^2 = \sum_a (a - \langle A \rangle)^2 P(a) = \langle\psi|\bar{A}^2|\psi\rangle$

**Shift invariance:**  $(\Delta A)^2 = \langle\bar{A}^2\rangle$ ,  $[A, \bar{B}] = [A, B]$

**Uncertainty:**  $\Delta A \Delta B \geq \frac{1}{2} |\langle\psi|[A, B]|\psi\rangle|$

## CLASSICAL CORRELATION

**Setup 1:** Charlie prepares system; Alice and Bob measure;

◦ Outcomes:  $\sigma = \pm 1$  (*penny* = -1, *dime* = +1)

◦ Means:  $\langle\sigma_A\rangle = \langle\sigma_B\rangle = 0$

◦ Correlation:  $\langle\sigma_A \sigma_B\rangle = -1$

◦ Relation:  $\langle\sigma_A \sigma_B\rangle \neq \langle\sigma_A\rangle \langle\sigma_B\rangle$

**Setup 2:** Charlie-A prepares system, Alice measure;

Charlie-B prepares system, Bob measure;

◦ Correlation = 0

**Probability:**  $P(a,b) = P(a)P(b)$  if  $a, b$  independent

**TENSOR PRODUCT:**  $\{S_a, S_b\}_{\text{basis}}$ ;  $\dim\{N_a, N_b\}$

$(S_a \otimes S_b)$ ;  $\dim(S_a \otimes S_b) = N_a N_b$

**States:**  $|\psi\rangle \in S_a$ ,  $|\phi\rangle \in S_b$ ; no cross-space addition

**Combined Basis:**  $\{|a_i b_j\rangle\} = \{|a_i\rangle \otimes |b_j\rangle\}$ ;

$\langle a b | a' b' \rangle = \delta_{aa'} \delta_{bb'}$ ;  $= 1 \iff a = a', b = b'$

**Combined State:**  $|\psi\rangle = \sum_{a,b} \psi(a,b) |ab\rangle$

## TWO SPIN SYSTEM:

$\{|u_j\rangle, |d_j\rangle\}_{\text{basis}} \mid j \in \{a, b\}$

**States:**  $|\psi_a\rangle = \alpha_u |u\rangle + \alpha_d |d\rangle$ ;  $|\psi_b\rangle = \beta_u |u\rangle + \beta_d |d\rangle$

**Product States: 4 DOF**

◦  $|\psi_{a,b}\rangle = \alpha_u \beta_u |uu\rangle + \alpha_u \beta_d |ud\rangle + \alpha_d \beta_u |du\rangle + \alpha_d \beta_d |dd\rangle$

◦  $|\psi_a\rangle \otimes |\psi_b\rangle \Rightarrow \sum_{ij} \alpha_i \beta_j |ij\rangle$ ; 8 real  $\rightarrow$  4 (norm + phase/spin)

◦ Minimal Parameterized State:  $|\psi_a\rangle = \cos \frac{\theta_a}{2} |u\rangle + e^{i\phi_a} \sin \frac{\theta_a}{2} |d\rangle$

**General State: 6 DOF**

◦  $|\psi_{a,b}\rangle = \psi_{uu} |uu\rangle + \psi_{ud} |ud\rangle + \psi_{du} |du\rangle + \psi_{dd} |dd\rangle$

◦  $|\psi\rangle = \sum_{ij} \psi_{ij} |ij\rangle$ ; 8 real  $\rightarrow$  6 (global norm + phase)

**Entangled States:**

Complete description of combined system

**Maximally Entangled States:**  $\{|singlet\rangle, |triplet(3D)\rangle\}$

◦ Nothing can be known about individual subsystems

◦  $|singlet\rangle = 1/\sqrt{2}(|ud\rangle - |du\rangle)$

$$\circ \begin{cases} |T_+\rangle = |uu\rangle \\ |T_0\rangle = \frac{1}{\sqrt{2}}(|ud\rangle + |du\rangle) \\ |T_-\rangle = |dd\rangle \end{cases} \quad |triplet(3D)\rangle$$

◦  $|singlet\rangle \oplus |triplet(3D)\rangle =$  full 4D space

**Observables:**  $\{\sigma_x, \sigma_y, \sigma_z\} \in S_a$ ;  $\{\tau_x, \tau_y, \tau_z\} \in S_b$

◦  $\sigma_z |u\rangle = |u\rangle$ ;  $\sigma_z |d\rangle = -|d\rangle$ ;

$\sigma_x |u\rangle = |d\rangle$ ;  $\sigma_x |d\rangle = |u\rangle$ ;  $\sigma_y |u\rangle = i|d\rangle$ ;  $\sigma_y |d\rangle = -i|u\rangle$ ;

$\tau_x |u\rangle = |u\rangle$ ;  $\tau_x |d\rangle = -|d\rangle$ ;

$\tau_y |u\rangle = i|d\rangle$ ;  $\tau_y |d\rangle = -i|u\rangle$ ;

$\sigma_x |du\rangle = (\sigma_x \otimes I)(|d\rangle \otimes |u\rangle)$ ;  $\tau_z |du\rangle = (I \otimes \tau_z)(|d\rangle \otimes |u\rangle)$

◦ In Product States, Expectation values of ind. components of

$\sigma, \tau$  remains same i.e., Spin Polarization Principle holds.

$\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1$

◦ In Maximally Entangled State: Expectation value is zero

$\langle sing|\sigma_x|sing\rangle = \langle sing|\sigma_y|sing\rangle = \langle sing|\sigma_z|sing\rangle = 0$

**Composite Observables:**

◦  $\tau_z \sigma_z |sing\rangle = -|sing\rangle$

◦  $\tau_x \sigma_x |sing\rangle = -|sing\rangle$  (Non Obvious)

◦  $\tau_z \sigma_z |T_0\rangle = -|T_0\rangle$

◦  $\tau_z \sigma_z |T_+\rangle = |T_+\rangle$

◦  $\tau_z \sigma_z |T_-\rangle = |T_-\rangle$

◦ Expectation values :  $i \in x, y, z$

| State | $\langle\sigma_i\rangle$ | $\langle\sigma_i \otimes \sigma_i\rangle$ |
|-------|--------------------------|-------------------------------------------|
| $T_+$ | (0, 0, 1)                | (0, 0, 1)                                 |
| $T_-$ | (0, 0, -1)               | (0, 0, 1)                                 |
| $T_0$ | (0, 0, 0)                | (1, 1, -1)                                |
| $S$   | (0, 0, 0)                | (-1, -1, -1)                              |

◦ The Combined Operator: Energy based measurement

$(\sigma \cdot \tau) = \sigma_x \otimes \tau_x + \sigma_y \otimes \tau_y + \sigma_z \otimes \tau_z$

$(\sigma \cdot \tau) |uu\rangle = |uu\rangle$

$(\sigma \cdot \tau) |dd\rangle = |dd\rangle$

$(\sigma \cdot \tau) |ud\rangle = 2|du\rangle - |ud\rangle$

$(\sigma \cdot \tau) |du\rangle = 2|ud\rangle - |du\rangle$

$(\sigma \cdot \tau) |T_i\rangle = +|T_i\rangle$  | All Triplets with +1 eigval

$(\sigma \cdot \tau) |S\rangle = -3|S\rangle$

## TENSOR PRODUCT OF MATRICES

**Kronecker Product:**  $A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{pmatrix}$

◦  $(A \otimes B)(a \otimes b) = (Aa \otimes Bb)$

**Outer Product:**  $|\psi\rangle\langle\phi|; (|\psi\rangle\langle\phi|)|A\rangle = |\psi\rangle\langle\phi|A\rangle$

**Projection Operator:**  $|\psi\rangle\langle\psi|$ ; Hermitian; eig: 1 (on  $|\psi\rangle$ ), 0 (orthog)

◦  $(|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|$ ;  $\text{Tr} = 1$ ;  $\sum_i |i\rangle\langle i| = I$

◦  $\text{Tr}|\psi\rangle\langle\psi|L = \langle\psi|L|\psi\rangle = \langle L \rangle$

## DENSITY MATRICES (OPERATORS)

When uncertain abt. prepared state:  $\rho$

**Pure State:**  $\langle L \rangle = \text{Tr}|\psi\rangle\langle\psi|L = \text{Tr}\rho L$

◦ Density Matrix is the Projection Operator

**Mixed State:**

$\langle L \rangle = 1/2 \text{Tr}|\psi\rangle\langle\psi|L + 1/2 \text{Tr}|\phi\rangle\langle\phi|L$

$\langle L \rangle = \text{Tr}1/2|\psi\rangle\langle\psi| + 1/2|\phi\rangle\langle\phi|L$

$\langle L \rangle = \text{Tr}\rho L$  |  $\rho = 1/2|\psi\rangle\langle\psi| + 1/2|\phi\rangle\langle\phi|$

◦ General Form:

$\rho = P_1|\phi_1\rangle\langle\phi_1| + P_2|\phi_2\rangle\langle\phi_2| + \dots$   $\langle L \rangle = \sum_{a,a'} L_{a',a} \rho_{a,a'}$

**Entanglement**

$A_{sys}, B_{sys}$ ; suppose we are interested in measurmt. of obser.

$L_a$  for  $A_{sys}$

$\langle L \rangle = \sum_{a'b',ab} \psi^*(a'b')L_{a'b',ab}\psi(ab)$

$\langle L \rangle = \sum_{a',b,a} \psi^*(a'b)L_{a',a}\psi(ab)$

$\rho_{aa'} = \sum_b \psi^*(a'b')\psi(ab)$

◦ **What are**  $L_{a',a}, \rho_{aa'}$ ?

- Alice's observable and density matrix while assuming

Bob's state doesn't change

◦ **Alice's state transitions, assuming const Bob's state**

-  $\rho_a(0'0)_{0'->0} = \rho_{a'b',ab}(00)'(00) + \rho_{a'b',ab}(01)'(01)$

-  $\rho_a(0'1)_{0'->1} = \rho_{a'b',ab}(00)'(10) + \rho_{a'b',ab}(01)'(11)$

-  $\rho_a(1'0)_{1'->0} = \rho_{a'b',ab}(10)'(00) + \rho_{a'b',ab}(11)'(01)$

-  $\rho_a(1'1)_{1'->1} = \rho_{a'b',ab}(10)'(10) + \rho_{a'b',ab}(11)'(11)$

- same steps for  $L_{a',a}$

◦ **Pure State DM** ( $\rho$ ) of a subsystem can be written as the projection operator of a state  $\psi$ ;

-  $\rho = |\psi\rangle\langle\psi|$ ;  $\rho^2 = \rho$ ,  $\text{Tr}(\rho^2) = 1$

◦ **Mixed state DM** ( $\rho$ ) of a subsystem can not be written in the form of a projection operator

-  $\rho_{mixed}$ ;  $\rho^2 \neq \rho$ ,  $\text{Tr}(\rho^2) < 1$

## TESTS FOR ENTANGLEMENT

**Correlation Test:**

◦ Given observables A, B and their Product State AB

a. Correlation  $C(A, B) = \langle AB \rangle - \langle A \rangle \langle B \rangle$

b. Sign indicates the correlation direction

**Density Matrix Test:**

◦ If we have the Composite System with one of the subsystem's Density Matrix  $\rho_{a'a}$

1. Compute the Eigen Values of the Density Matrix

a. Pure State: eig vals = {1, zeroes}

b. Mixed State: eig vals = {sum to 1}

c. Max Entangled State:  $\rho_{a'a} = aI$ ; eig.vals =  $\{1/N_a\}$

2. Pure State  $\implies$  Complete Subsystem information

3. Max.Ent State

$\implies$  Zero information abt. own's Subsystem

$\implies$  Can predict behaviour of other Subsystems

## NON LOCALITY (Faster than Speed of Light)

Entanglement leads to non locality. But, local subsystem's

unitary evolutions doesn't effect the remote subsystem's

Density Matrix (avoids faster than speed of light problem).

Derived from *Quantum Mechanics: The Theoretical Minimum*  
Leonard Susskind & Art Friedman